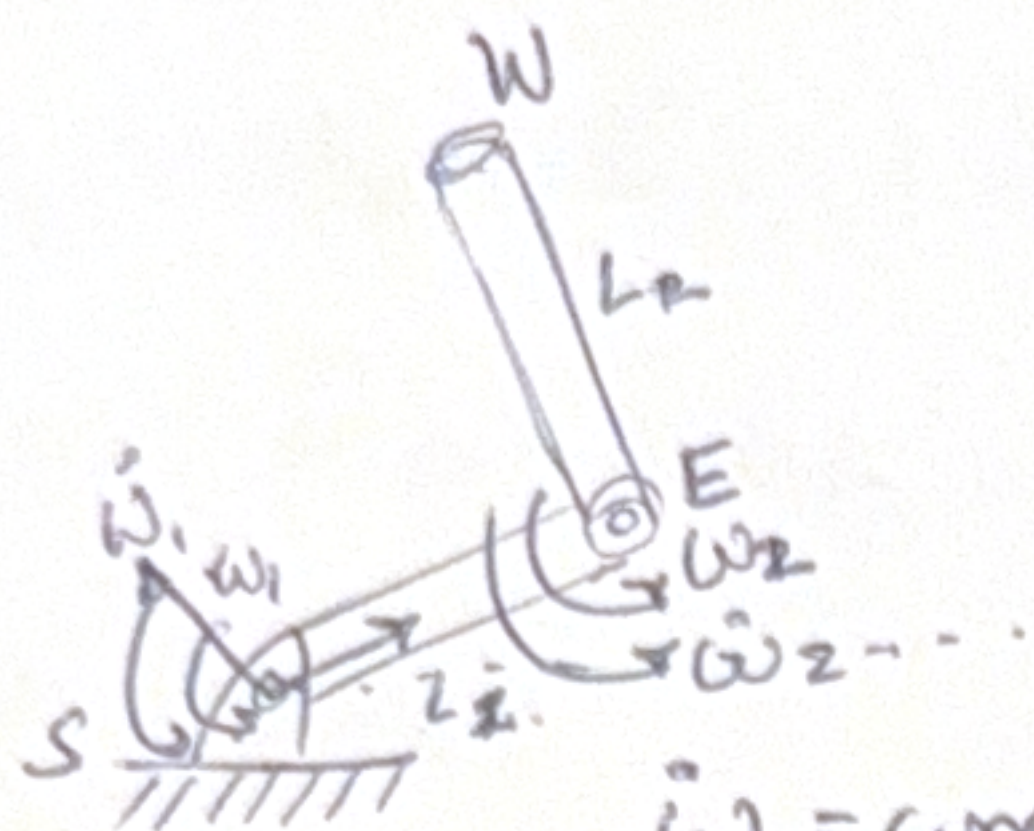




$$\begin{aligned}\vec{R}_{W/O} &= \vec{R}_{E/O} + \vec{R}_{W/E} \\ &= \vec{R}_{S/O} + \vec{R}_{E/S} + \vec{R}_{W/E} \\ &= \vec{R}_{E/S} + \vec{R}_{W/E}.\end{aligned}$$

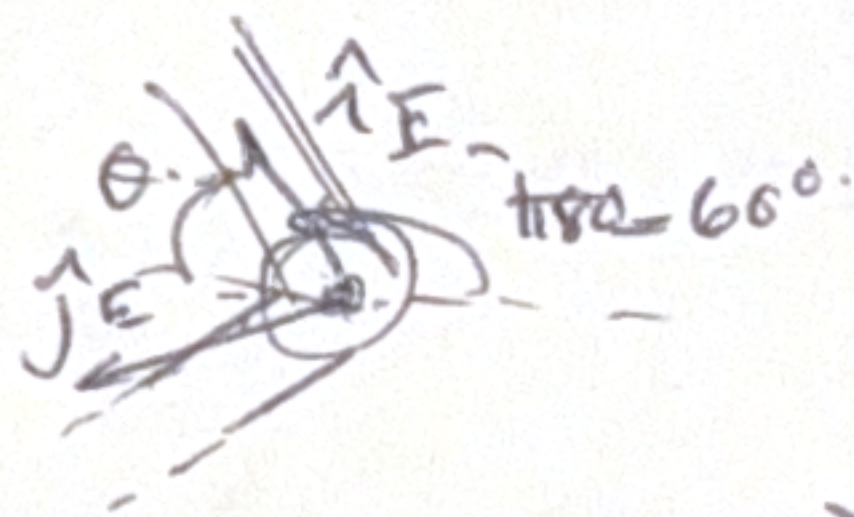
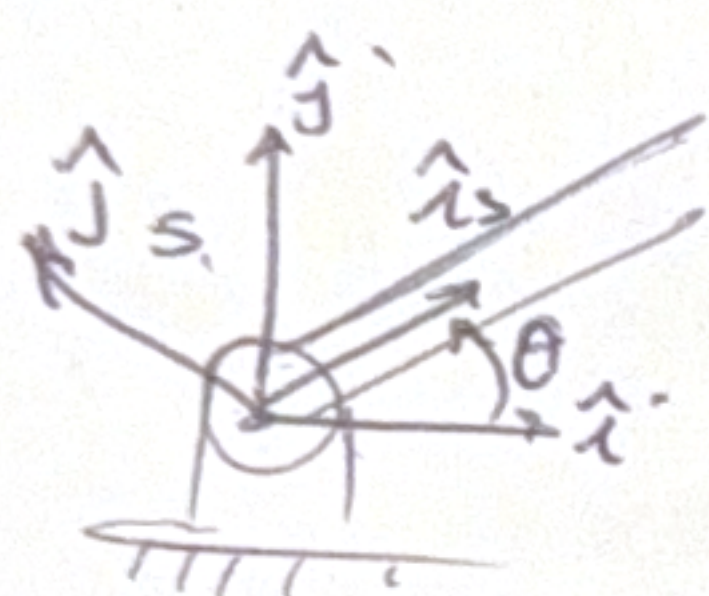
* fixed shoulder pin
free to rotate

$\vec{\omega}_1 = \text{const}$
 $\vec{\omega}_2 = \text{const}$
 $\vec{\omega}_2$ relative to $\vec{\omega}_1$
 $\vec{\omega}_2$ relative to $\vec{\omega}_1$

$$\frac{d\vec{R}_{W/O}}{dt} = \vec{V}_{W/O} = \vec{V}_{E/O} + (\vec{V}_{W/E})_{\text{rel}} + \vec{\omega}_W \times \vec{R}_{W/E}$$

$$= \vec{V}_{S/O} + (\vec{V}_{E/S})_{\text{rel}} + \vec{\omega}_E \times \vec{R}_{E/S} + (\vec{V}_{W/E})_{\text{rel}} + \vec{\omega}_W \times \vec{R}_{W/E}$$

Fix O frame on the shoulder ~~pin~~ pin.



note $\vec{V}_{S/O} = (\vec{V}_{E/S})_{\text{rel}} = (\vec{V}_{W/E})_{\text{rel}} = 0$
due to fixed arm lengths and fixed shoulder pin.

$$\vec{V}_{W/O} = \vec{\omega}_E \times \vec{R}_{E/S} + \vec{\omega}_W \times \vec{R}_{W/E}$$

$$\vec{a}_{W/O} = \vec{a}_{E/O} + (\vec{a}_{W/E})_{\text{rel}} + 2\vec{\omega}_W \times (\vec{V}_{W/E})_{\text{rel}} + \vec{\omega}_W \times \vec{R}_{W/E} + \vec{\omega}_W \times (\vec{\omega}_W \times \vec{R}_{W/E})$$

since $(\vec{V}_{W/E})_{\text{rel}} = 0$, $(\vec{a}_{W/E})_{\text{rel}} = 0$ again due to fixed arm lengths.

$$\vec{a}_{W/O} = \vec{a}_{E/O} + \vec{\omega}_W \times \vec{R}_{W/E} + \vec{\omega}_W \times (\vec{\omega}_W \times \vec{R}_{W/E})$$

$$\begin{aligned}\vec{a}_{E/O} &= \vec{a}_{S/O} + (\vec{a}_{E/S})_{\text{rel}} + 2\vec{\omega}_E \times (\vec{V}_{E/S})_{\text{rel}} + \vec{\omega}_E \times \vec{R}_{E/S} \\ &= \vec{\omega}_E \times \vec{R}_{E/S} + \vec{\omega}_E \times (\vec{\omega}_E \times \vec{R}_{E/S}) + \vec{\omega}_E \times (\vec{\omega}_E \times \vec{R}_{E/S})\end{aligned}$$

$$\vec{a}_{W/O} = \vec{\omega}_E \times \vec{R}_{E/S} + \vec{\omega}_E \times (\vec{\omega}_E \times \vec{R}_{E/S}) + \vec{\omega}_W \times \vec{R}_{W/E} + \vec{\omega}_W \times (\vec{\omega}_W \times \vec{R}_{W/E})$$

Now, we need expressions for each term.

first, observe that $\hat{k} = \hat{k}_E = \hat{k}_S$

$$\vec{R}_{E/S} = L_1 \hat{i}_S, \vec{R}_{W/E} = L_2 \hat{i}_E$$

$$\vec{\omega}_E = \vec{\omega}_1 = \omega_1 \hat{k}_S, \vec{\omega}_W = (\omega_1 + \omega_2) \hat{k}_E$$

$$\vec{\omega}_E = \vec{\omega}_1, \vec{\omega}_W = \vec{\omega}_1 + \vec{\omega}_2 + \vec{\omega}_1 \times \vec{\omega}_2$$

since both in $\hat{k}$ direction.

$$\vec{\omega}_W = \omega_1 \hat{k} + \omega_2 \hat{k}$$

Rewritten:

$$\vec{a}_{W/O} = \vec{\omega}_E \times \vec{R}_{E/S} + \vec{\omega}_E \times (\vec{\omega}_E \times \vec{R}_{E/S}) + \vec{\omega}_W \times \vec{R}_{W/E} + \vec{\omega}_W \times (\vec{\omega}_W \times \vec{R}_{W/E})$$

$$\vec{V}_{W/O} = \vec{\omega}_E \times \vec{R}_{E/S} + \vec{\omega}_W \times \vec{R}_{W/E}$$

and define

$$\vec{R}_{E/S} = L_1 \hat{i}_S, \vec{R}_{W/E} = L_2 \hat{i}_E.$$

$$\vec{\omega}_E = \omega_1 \hat{k}_S, \vec{\omega}_W = (\omega_1 + \omega_2) \hat{k}_E.$$

$$\begin{aligned} \vec{\omega}_E &= \dot{\omega}_1 \hat{k}_S, \vec{\omega}_W = \vec{\omega}_1 + \vec{\omega}_2 \\ &= \dot{\omega}_1 \hat{k} + \dot{\omega}_2 \hat{k} \\ &= (\dot{\omega}_1 + \dot{\omega}_2) \hat{k} \end{aligned}$$

$$\text{note } \hat{k}_S = \hat{k}_E = \hat{k}.$$

$$\hat{i}_S = \begin{pmatrix} \cos \theta_S & -\sin \theta_S \\ \sin \theta_S & \cos \theta_S \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos \theta_S \\ \sin \theta_S \end{pmatrix} = \cos \theta_S \hat{i} + \sin \theta_S \hat{j}$$

$$\hat{i}_E = \begin{pmatrix} \cos \theta_E & -\sin \theta_E \\ \sin \theta_E & \cos \theta_E \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos \theta_E \\ \sin \theta_E \end{pmatrix} = \cos \theta_E \hat{i} + \sin \theta_E \hat{j}$$

$$\hat{j}_S = \begin{pmatrix} \cos \theta_S & -\sin \theta_S \\ \sin \theta_S & \cos \theta_S \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin \theta_S \\ \cos \theta_S \end{pmatrix} = -\sin \theta_S \hat{i} + \cos \theta_S \hat{j}$$

$$\hat{j}_E = \begin{pmatrix} \cos \theta_E & -\sin \theta_E \\ \sin \theta_E & \cos \theta_E \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin \theta_E \\ \cos \theta_E \end{pmatrix} = -\sin \theta_E \hat{i} + \cos \theta_E \hat{j}$$

note

$$\vec{\omega}_E \times \vec{R}_{E/S} = \omega_1 \hat{k}_S \times L_1 \hat{i}_S = L_1 \omega_1 \hat{j}_S$$

$$\vec{\omega}_W \times \vec{R}_{W/E} = (\omega_1 + \omega_2) \hat{k}_E \times L_2 \hat{i}_E = L_2 (\omega_1 + \omega_2) \hat{j}_E$$

$$\vec{\omega}_E \times (\vec{\omega}_E \times \vec{R}_{E/S}) = \vec{\omega}_E \times (L_1 \omega_1 \hat{j}_S) = \omega_1 \hat{k}_S \times (L_1 \omega_1 \hat{j}_S) = L_1 \omega_1^2 (-\hat{i}_S)$$

$$\vec{\omega}_W \times (\vec{\omega}_W \times \vec{R}_{W/E}) = \vec{\omega}_W \times ((\omega_1 + \omega_2) L_2 \hat{j}_E) = (\omega_1 + \omega_2) \hat{k}_E \times (L_2 (\omega_1 + \omega_2) \hat{j}_E)$$

$$\vec{\omega}_E \times \vec{R}_{E/S} = \dot{\omega}_1 \hat{k}_S \times L_1 \hat{i}_S = L_1 \dot{\omega}_1 \hat{j}_S$$

$$\vec{\omega}_W \times \vec{R}_{W/E} = (\dot{\omega}_1 + \dot{\omega}_2) \hat{k}_S \times L_2 \hat{i}_E = L_2 (\dot{\omega}_1 + \dot{\omega}_2) \hat{j}_E$$

$$\vec{V}_{W/O} = L_1 \dot{\omega}_1 \hat{j}_S + L_2 (\dot{\omega}_1 + \dot{\omega}_2) \hat{j}_E$$

$$\vec{a}_{W/O} = L_1 \dot{\omega}_1 \hat{j}_S + L_1 \omega_1^2 (-\hat{i}_S) + L_2 (\dot{\omega}_1 + \dot{\omega}_2) \hat{j}_E + L_2 (\omega_1 + \omega_2)^2 (-\hat{i}_E).$$